



Technical Note

Spatial and seasonal variations in atrazine and metolachlor surface water concentrations in Ontario (Canada) using ELISA

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ABSTRACT

Concerns regarding the impacts of pesticides on aquatic species and drinking water sources have increased demands on water quality monitoring programs; however the costs of sample analysis can be prohibitive. In this study we investigated enzyme-linked immunosorbent assay (ELISA) as a cost-effective, high through-put method for measuring pesticide concentrations in surface waters. Seven hundred and thirty-nine samples from 158 locations throughout Ontario were analysed for atrazine and metolachlor from April to October 2007. Concentrations ranged from <0.1 to 3.91 $\mu\text{g L}^{-1}$ (median = 0.12 $\mu\text{g L}^{-1}$) for atrazine and from <0.1 to 1.83 $\mu\text{g L}^{-1}$ (median = 0.09 $\mu\text{g L}^{-1}$) for metolachlor. Peak concentrations occurred in late spring/early summer, in rural agricultural locations, and decreased over the remainder of the growing season for both herbicides. About 3% of the samples that had ELISA results occurring above the limit of quantification (0.10 $\mu\text{g L}^{-1}$) were evaluated against gas chromatography-mass spectrometry (GC-MS). Linear regression analysis revealed a R^2 value of 0.88 and 0.39, for atrazine and metolachlor, respectively. ELISA tended to overestimate concentrations for atrazine and metolachlor, most likely because the ELISA kits also detect their metabolites. Atrazine data suggest that ELISA may be used complementary with GC-MS analysis to enhance the spatial and temporal resolution of a water quality monitoring study. The commercially available metolachlor ELISA kit requires further investigation. ELISA may be used to detect atrazine and metolachlor in surface water samples, but it is not recommended as a quantitative replacement for traditional analytical methods.

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1. Introduction

Atrazine (2-chloro-4-ethylamino-6-isopropylamino-s-triazine) and metolachlor [2-chloro-N-(2-ethyl-6-methylphenyl)-N-(2-methoxy-1-methylethyl) acetamide] are two of the most widely used herbicides in Ontario. In 2008, about 460 000 kg of atrazine and 472 000 kg of metolachlor were applied in Ontario for agricultural use (McGee et al., 2010). They are most often applied pre-emergence (mid-April to mid-May) or post-emergence (end of June) to control broadleaf and annual grassy weeds in corn and bean fields (Extoxnet, 2007a,b). These herbicides can enter surface waters through a range of transport mechanisms and can affect water quality. Their occurrence in surface waters depends on a variety of factors such as application rate, persistence and mobility, rainfall, topography, and climate (Lin et al., 1999).

Atrazine and metolachlor are persistent in soils with a reported half-life range of 15 to >60 d depending on soil composition such as fulvic acid and soil microbial activity (Rhoton et al., 2002; Extoxnet, 2007a,b). They are moderately water soluble (30 and 530 mg L⁻¹,

respectively), may be flushed from soils during heavy rains common during spring and fall runoff in temperate climates, and are often detected in runoff water and groundwater in high use areas (Thurman et al., 1991; Selim, 2003). In southwestern Ontario, values as high as 95 $\mu\text{g L}^{-1}$ have been detected in tributaries draining agricultural areas, and in corn growing areas of the Midwestern US, values as high as 250 $\mu\text{g L}^{-1}$ have been detected in field drains (Takacs et al., 2002). Metolachlor concentrations as high as 97 $\mu\text{g L}^{-1}$ were reported in Ohio tributaries to Lake Erie (Richards and Baker, 1993) and as high as 293 $\mu\text{g L}^{-1}$ in surface runoff from a southern Ontario cornfield (Buttle, 1990). Studies show that at environmentally relevant concentrations atrazine and metolachlor may pose potential health hazards to vertebrate and invertebrate populations (Dutka et al., 1995; Hayes et al., 2002; Osano et al., 2002; Waring and Moore, 2004; Brod-kin et al., 2007; Suzawa and Ingraham, 2008).

In Ontario, almost all (>99%) of herbicides used occur within the Great Lakes/St. Lawrence watershed (McGee et al., 2010). Concentration data are available for the Great Lakes (Struger et al., 2004; Kannan et al., 2006; Kurt-Karakus et al., 2010) and select streams and tributaries in Ontario (Frank et al., 1990, 1991; Gaynor et al., 1998; Struger and Fletcher, 2007; Kurt-Karakus et al., 2008). Although, limited seasonal data exist and no information is available on a provincial scale.

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Most monitoring programs have budgetary constraints which generally limit the number of samples analysed, compromising either the spatial or temporal resolution. Cost-effectiveness is important to optimize monitoring budgets and maximize information as long as there is no significant reduction in the accuracy and sensitivity of analysis. Enzyme-linked immunosorbent assay (ELISA) has been used for cost-effective and fast turn-around-time analyses with little to no sample pretreatment (Rubio et al., 1991; Hall et al., 1993; Lawruk et al., 1994; Zaruk et al., 1998; Graziano et al., 2006; Byer et al., 2008b). Previous studies have compared ELISA to conventional gas chromatography-mass spectrometry (GC-MS) (Schraer et al., 2000; Zaruk et al., 2001; Adams et al., 2004; Kramer et al., 2007). These studies indicate that ELISA is highly reproducible, but potential biases and interferences may cause inaccuracies in ELISA test results related to cross-reactivity. In this study we evaluate the occurrence of the current-use herbicides atrazine and metolachlor in Ontario surface waters using ELISA, and show some of the advantages of ELISA for a monitoring program by making use of a complementary study designed for increased spatial and temporal resolution.

2. Materials and methods

2.1. Sample collection

Seven hundred and thirty-nine samples from 158 sites across Ontario were collected monthly from April to October 2007. All sampling locations occurred within the Great Lakes/St. Lawrence Watershed (Supplementary material (SM), Figs. SM-1, SM-2, and Table SM-1). A subset of samples that had ELISA results greater than the limit of quantification (LOQ) of $0.1 \mu\text{g L}^{-1}$ for atrazine and/or metolachlor were analysed with GC-MS for comparative purposes. Samples for comparative studies were collected by grab sampling in 1 L amber glass Boston round bottles (Fisher Scientific, Ottawa, ON), a few cm below the surface. Subsamples (10 mL) were collected in 20 mL clear glass vials (Fisher Scientific, Mississauga, ON) and frozen on their side between -16 and -22°C until analysis with ELISA. The remainder of the 1 L samples were submitted for neutral herbicide analysis to the National Laboratory for Environmental Testing (NLET, Environment Canada, Burlington, ON). All other samples were collected in 20 mL clear glass vials and stored as described above. The ELISA samples were stored for <4 wk after collection until analysis (Byer et al., 2008a), and were filtered using Nylon, non-sterile $0.2 \mu\text{m}$ syringe filters and Norm-jet 20 mL syringes (Fisher Scientific, Mississauga, ON) to remove suspended materials and reduce the effects of sediment. Of the 739 samples, 210 and 529 were collected from sites classified as urban and rural, respectively. Sites were considered urban if the sampling location occurred within a city with a population >10 000 people. All other sample locations were considered rural. Twenty percent of samples were analysed in duplicate with ELISA.

2.2. Analysis

2.2.1. ELISA

RaPID Assay ELISA test kits were obtained for atrazine and metolachlor from Osprey Scientific (Mississauga, ON). The analysis procedure was performed in accordance with the operating data-sheet accompanying the test kits. Vortexing was accomplished using a Vortex Genie 2 (VWR International; Edmonton, AB), magnetic separation consisted of a test tube holder fitted over a magnetic separator rack, and analysis was performed with a SDI RPA-II Photometric Analyzer (Osprey Scientific Inc., Edmonton, AB).

2.2.2. Conventional analysis

Neutral herbicide analysis of surface water samples was performed at NLET. A detailed description of the analytical method has been previously outlined (Donald et al., 2007). Extraction efficiency was monitored by the addition of d14-trifluralin prior to sample analysis. Surface water samples (1.0 L) were extracted with dichloromethane (DCM), concentrated and solvent exchanged into isooctane. Target compounds were purified using modified Florisil and collected with 2% methanol in DCM before being concentrated to 1 mL prior to instrumental analysis. Extracts were analysed by GC-MS using an Agilent 6890 GC interfaced with a 5973 mass selective detector by electron ionization in selected ion monitoring mode. The LOQ for metolachlor was 24 ng L^{-1} , and 5.8 ng L^{-1} for atrazine.

2.2.3. Data analysis

The Student's *t*-test was used to test for differences in mean concentrations of atrazine and metolachlor by land use (urban vs. rural), region and season. Non-detectable concentrations were assigned a value of half the detection limit for calculating arithmetic means. Relationships between ELISA and GC-MS results were tested using simple linear regression. Tests were considered significant if the *p* value was <0.05.

2.3. Quality control

Descriptions of the quality control measures, including instrumental variability and ELISA analysis, were previously detailed (Byer et al., 2008b). Calibration using a three-point curve and a mid-range standard (control) for each pesticide averaged over 30 determinations for each pesticide (Table 1) gave a $R^2 > 0.990$. The operating manuals for the test kits specify a 20% standard deviation (SD) for the control, which was more than adequate for atrazine (mean of 7.5%), with no individual SD results greater than 20%. The metolachlor test showed more variability (mean of 22%); with individual control SD results occurring outside of the acceptable range 37% of the time. All results for metolachlor recorded when the control deviated by greater than 20% were adjusted to the control by multiplying the actual result by the factor (*f*):

$$f = 1 + \frac{(\text{actual} - \text{theoretical})}{\text{theoretical}} \quad (1)$$

3. Results and discussion

3.1. Provincial water quality monitoring

We analysed 739 grab samples from various surface waters throughout Ontario for atrazine and metolachlor using ELISA. Sample timing was approximately monthly, although the collection date was mostly random. Some samples were collected during or following rain events; therefore, the data represent a wide range of surface water conditions from low to high flow. We observed higher concentrations (up to 10-fold) in samples that were collected during a rain event compared to low flow conditions. This observation agrees with our previous results for glyphosate (Byer et al., 2008b) and with other reports which attribute high concentrations to flushing events (Thurman et al., 1991; Graziano et al., 2006) or rain initiated, high surface water runoff.

3.1.1. Environmental trends

The occurrence of atrazine and metolachlor in surface waters was not confined to a specific geographic area (Figs. 1 and SM-3). Both herbicides were, for the most part, detected in every region monitored. However, peak concentrations were measured in rural,

Table 1
ELISA calibration and control data, $\mu\text{g L}^{-1}$ ^a.

	Equation of line	R^2	Experimental	Theoretical	% Diff.
Atrazine	$y = -0.8493x + 0.027$	0.997	2.9 ± 0.2	3.0 ± 0.6	(16) to 4
Metolachlor	$y = -0.8144x + 0.193$	0.999	2.9 ± 0.7	3.0 ± 0.6	(52) to 35

^a Mean value of 30 calibrations.

agricultural locations, in southwestern Ontario (Table 2), corresponding to rain events. Lower levels were measured in urban centers. Concentrations of both herbicides were significantly higher in southern Ontario in relation to the northern region throughout our study ($p < 0.001$). Across southern Ontario the mean atrazine concentration ($0.28 \mu\text{g L}^{-1}$) was higher than metolachlor ($0.20 \mu\text{g L}^{-1}$); however, for northern Ontario the difference was not significant ($p > 0.7$). The data correspond well with known application practices given that the majority of agricultural activities in Ontario occur in the southern region, comprising about 95% (3.78 of 3.98 million ha) of the total arable land (Statistics Canada, 2006). Also, the growing season and period of pesticide application in southern Ontario is several weeks longer than in northern Ontario.

The temporal trend data for atrazine and metolachlor were divided into primarily urban and rural influenced locations. Table 2 demonstrates rural samples having elevated levels compared to ur-

ban samples, which is predictable, since atrazine and metolachlor are used exclusively for agriculture in Ontario (Takacs et al., 2002). The detection frequency (DF) was higher for samples from rural surface waters 54% (286 of 534) compared with urban streams 36% (73 of 205) for atrazine, and the mean concentration was higher for rural samples ($0.27 \mu\text{g L}^{-1}$) compared to urban samples ($0.17 \mu\text{g L}^{-1}$) ($p < 0.01$). The difference in DF for metolachlor was less significant for rural 37% (198 of 534) vs. urban 33% (68 of 205) samples; however, the mean concentration for rural samples ($0.19 \mu\text{g L}^{-1}$) compared to urban samples ($0.10 \mu\text{g L}^{-1}$) remained significantly higher ($p < 0.001$). Our results for atrazine and metolachlor are in agreement with the DF and concentration range reported for Ontario streams (Zaruk et al., 1998; Kurt-Karakus et al., 2008).

Our data collection methodology may have implications for determining the potential impacts of agriculture activities down-

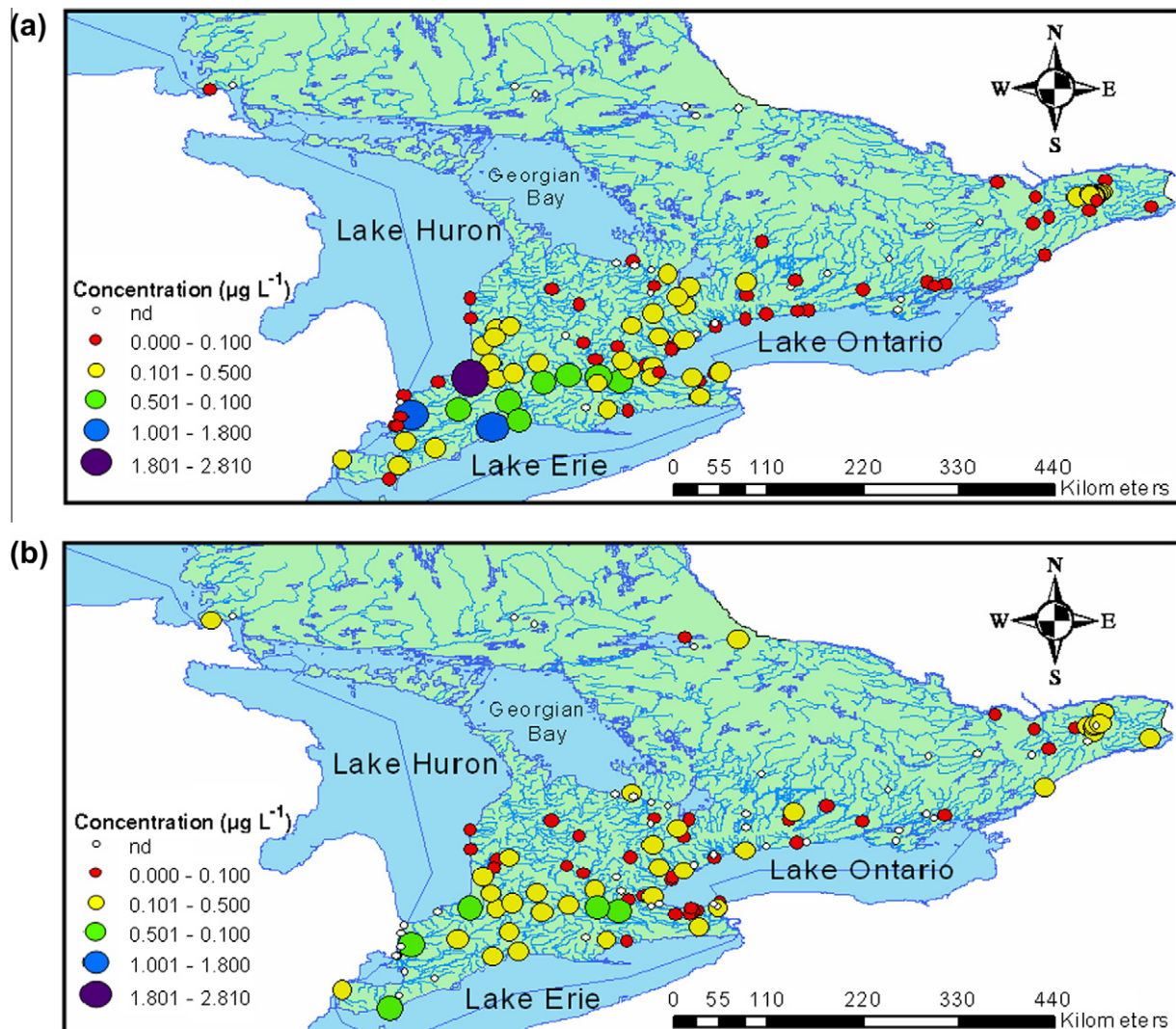


Fig. 1. Mean atrazine (a) and metolachlor (b) surface water concentration in the spring 2007.

Table 2
Atrazine and metolachlor concentrations ($\mu\text{g L}^{-1}$) in Ontario surface waters (2007).

	Min.	Max.	Mean	Median	SD	DF
<i>Atrazine</i>						
Urban	ND ^a	1.51	0.17	0.11	0.24	73/205
Rural	ND	3.91	0.27	0.14	0.47	286/534
North	ND	0.23	0.08	0.05	0.08	12/58
Southwest	ND	3.91	0.30	0.12	0.53	208/321
South-central	ND	1.51	0.17	0.11	0.23	51/158
Southeast	ND	0.91	0.21	0.13	0.21	88/202
Spring	ND	3.91	0.27	0.16	0.45	177/259
Summer	ND	3.34	0.28	0.15	0.43	100/236
Fall	ND	2.81	0.18	0.09	0.38	82/244
Total	ND	3.91	0.25	0.12	0.43	359/739
<i>Metolachlor</i>						
Urban	ND	0.54	0.10	0.06	0.10	68/205
Rural	ND	1.83	0.19	0.12	0.26	198/534
North	ND	0.22	0.07	0.05	0.06	13/58
Southwest	ND	1.83	0.22	0.12	0.30	134/321
South-central	ND	0.60	0.13	0.08	0.13	47/158
Southeast	ND	0.39	0.12	0.08	0.10	72/202
Spring	ND	1.61	0.21	0.14	0.25	119/259
Summer	ND	1.83	0.19	0.11	0.27	76/236
Fall	ND	0.45	0.08	0.04	0.09	71/244
Total	ND	1.83	0.17	0.09	0.23	266/739

^a ND represents non-detectable levels. Concentrations reported below the MDL ($0.05 \mu\text{g L}^{-1}$) were above the run detection limit ($>3 \times \text{blank}$); thus, these data should be regarded as trace levels.

stream on urban areas, especially when considering that atrazine was measured in an urban location as high as $1.51 \mu\text{g L}^{-1}$ and metolachlor upwards of $0.54 \mu\text{g L}^{-1}$. Struger and Fletcher (2007) have documented this trend for a variety of pesticides in primarily urban locations. The data show that when moderately persistent pesticides are applied in large enough quantities and in a variety of locations, their signature can be monitored a substantial distance from the source.

Table 2 and Fig. 2 also show the seasonality of atrazine and metolachlor surface water concentrations in Ontario. In general, there is a sharp rise in concentration in the spring (April–May) before peaking in mid-June, and steadily declining over the summer and into the fall. The observed trend corresponds well with known agricultural practices and precipitation patterns in Ontario (Fig. SM-4). Atrazine and metolachlor are typically applied in the spring pre-plant emergence to corn and soy, which requires little to no cultivation or reapplication after planting (Takacs et al., 2002; Extoxnet, 2007a,b). Accumulated rainfall is greatest in the spring/early summer in Ontario (Agriculture and Agri-Food Canada, 2007). As a result, one would expect peak concentrations in surface waters to occur within this time frame. Furthermore, precipitation “flush” events are strongest during the first storm after application and weaken with each subsequent storm. This results in declining herbicide concentrations in run-off with time (Graziano et al., 2006), which corresponds well with the observed trend. During this study we recorded six instances of a Canadian Council of Ministers of the Environment (CCME) Water Quality Guideline exceedance for atrazine, and several other results which occurred in close proximity to the guideline. These results were not confirmed by GC–MS. All of the exceedances occurred in southwestern Ontario in areas where atrazine levels have been elevated historically (Takacs et al., 2002), and usage has increased or become common practice (McGee et al., 2010).

3.1.2. GC–MS and ELISA comparison

We subjected a subset of 81 samples to our approach previously reported (Byer et al., 2008b). Fig. 3 and Table SM-2 show the GC–MS and ELISA comparison for all samples (24 of 81) which had measurable levels of atrazine or metolachlor above the ELISA

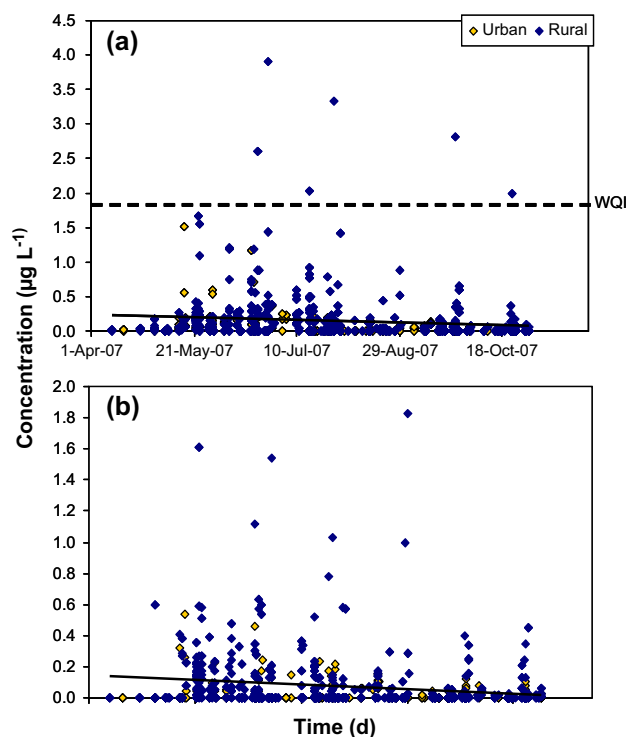


Fig. 2. Urban and rural, atrazine (a) and metolachlor (b) concentrations over the growing season 2007. The dotted-line (a) is the CCME water quality index (WQI) guideline for atrazine, and the solid-lines are data trend lines.

LOQ. Linear regression analysis for atrazine resulted in a R^2 of 0.88. This result is comparable to other findings (Hall et al., 1993; Graziano et al., 2006). The slope of the line indicates an over-estimation by ELISA which may be explained by matrix cross-related analogues (Rubio et al., 1991). The manufacturer of the ELISA kit lists several possible triazine herbicides which can confound the accuracy of detection and measurement of these herbicides (Table SM-3). GC–MS analyses confirmed the presence of these compounds namely, simazine, deethyl-atrazine (d-atrazine), cyanazine, and ametryn. Atrazine's main metabolite, d-atrazine, was added to the GC–MS comparison, and shifted the slope of the line from 0.83 to 1.02 without effecting the linear regression, R^2 . These data show a much closer agreement between the techniques, possibly indicating that the atrazine ELISA test kit detects atrazine and its metabolites.

Linear regression analysis for metolachlor resulted in a R^2 of 0.39; there was one instance of a false positive during a rain event at an urban creek. Our comparison data are in contrast with a previous study which reported a strong correlation for precipitation samples (Hall et al., 1993), but is in agreement with Schraer et al. (2000) who observed a similar poor agreement. The difference in linear regression may be reflective of the relatively low concentrations of metolachlor measured in this study. The metolachlor levels were less than atrazine in all of the samples analysed for comparison. The concentration of atrazine was on average 5-fold greater than metolachlor which is consistent with other reports (Struger et al., 2004). This is important considering the reported linear range of the kits is between 0.1 and $5.0 \mu\text{g L}^{-1}$. The metolachlor data were between 0.1 and $0.4 \mu\text{g L}^{-1}$; whereas, the atrazine data ranged from 0.1 to $1.6 \mu\text{g L}^{-1}$. The atrazine results extend further into the linear range of the calibration curve and may better reflect the true capability of the technique. The manufacturer also lists several possible cross-reactive analogues for metolachlor including, but not limited to, acetochlor, metalaxyl, and alachlor (Table SM-4). In addition, two common degradates metolachlor-

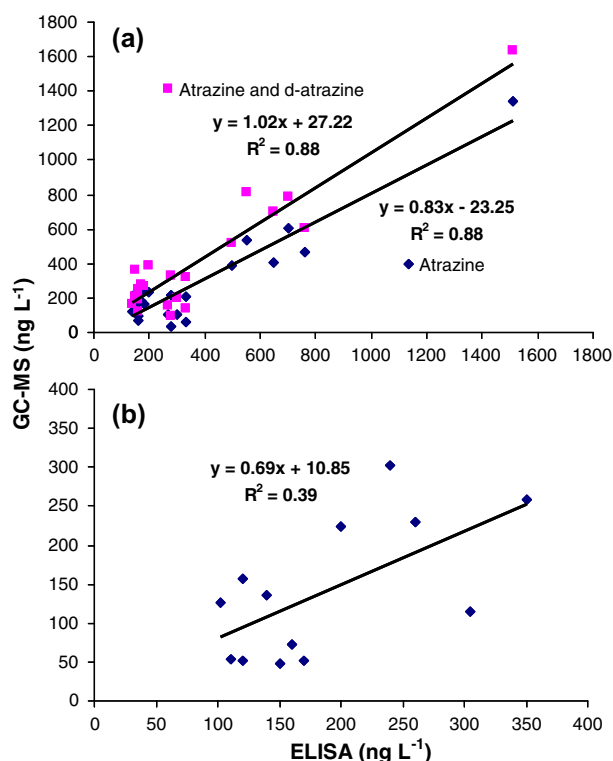


Fig. 3. Linear regression analysis of atrazine (a) and metolachlor (b) containing samples which occurred above 100 ng L⁻¹ using ELISA; $n = 23$ and $n = 13$ for atrazine and metolachlor, respectively. The $R^2 = 0.74$ for atrazine with the highest value removed.

ethanesulfonic acid and metolachlor-oxanilic acid, which are not present in precipitation but persist in soils, have been detected in surface water runoff at concentrations several times higher than metolachlor (Phillips et al., 1999; Schraer et al., 2000). These metolachlor metabolites were not analysed by the GC–MS method in this study, but may explain the apparent high bias of the metolachlor ELISA test kit, particularly the false positive that occurred during a rain event.

Concentrations of atrazine and metolachlor in the surface water samples analysed by our comparative study suggest that the atrazine ELISA kit may be a suitable complementary monitoring tool for increased spatial and temporal resolution; however, the metolachlor ELISA kit requires further investigation. Then again, one could use the atrazine ELISA kit as a pre-screening tool for both herbicides given that both herbicides are detected by GC–MS in the neutral herbicide screen. Furthermore, the CCME Water Quality Guideline for the protection of freshwater aquatic life is more than 4-fold less sensitive for metolachlor (CCME, 2001) based largely on related environmental effects.

4. Conclusions

The usefulness of ELISA as a method for cost-effective and fast turn-around-time analysis has been well established. However, its usefulness as a tool for quantitative analysis remains a limiting factor to its utility. We have shown, with varied outcomes, the advantage of using ELISA as a complementary analysis tool to enhance the spatial and temporal resolution of a pesticide monitoring program. It is worth noting that both ELISA test kits overestimated the pesticide concentrations when compared with GC–MS, most likely because the kits detect common metabolites as well as the target compound, and results occurring near the detection limit

had a high degree of variability. It is essential to investigate the suitability of an ELISA to a particular application prior to utilization in a broad-scale study design. Also, if ELISA kits are used for regulatory purposes, it is recommended that any results occurring within a defined safety factor of the guideline be confirmed with other traditional analytical methods.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.chemosphere.2010.12.054](https://doi.org/10.1016/j.chemosphere.2010.12.054).

References

- Adams, C., Jiang, H., McGuire, M., Graziano, N., Roberson, A., Frey, M., 2004. Accuracy and interferences for enzyme-linked immunoassay tests for atrazine. *J. Am. Water Works Assoc.* 96 (12), 126–139.
- Agriculture and Agri-Food Canada, 2007. National Agroclimate Information Service (NAIS), Ottawa, ON.
- Brodtkin, M.A., Madhoun, H., Rameswaran, M., Vatnick, I., 2007. Atrazine is an immune disruptor in adult northern leopard frogs (*Rana pipiens*). *Environ. Toxicol. Chem.* 26, 80–84.
- Buttle, J.M., 1990. Metolachlor transport in surface runoff. *J. Environ. Qual.* 19, 531–538.
- Byer, J.D., Struger, J., Klawunn, P., Sverko, E., 2008a. A Comprehensive Study of 2,4-D, Atrazine, Glyphosate and Metolachlor Using the ELISA Method for the CCME Water Quality Index. Water Science and Technology Directorate. Environment Canada, Burlington, ON.
- Byer, J.D., Struger, J., Klawunn, P., Todd, A., Sverko, E., 2008. Low cost monitoring of glyphosate in surface waters using the ELISA method: an evaluation. *Environ. Sci. Technol.* 42, 6052–6057.
- Canadian Council of Ministers of the Environment, 2001. Canadian Water Quality Guidelines for the Protection of Aquatic Life: CCME Water Quality Index 1.0, Technical Report. Canadian Environmental Quality Guidelines, 1999. CCME, Winnipeg, Manitoba, Canada.
- Donald, D.B., Cessna, A.J., Sverko, E., Glozier, E., 2007. Pesticides in surface drinking-water supplies of the northern Great Plains. *Environ. Health Perspect.* 11, 1183–1191.
- Dutka, B.J., McInnis, R., Pacepavicius, G.J., Maguire, F.J., 1995. Acute and Chronic Toxicity of the Herbicide Metolachlor to the Water Flea *Daphnia magna* and the Soil Nematode *Panagrellus redivivus*. National Water Research Institute, Burlington, ON.
- Extotoxnet, 2007a. Extension Toxicology Network: Pesticide Information Profiles–Atrazine. Oregon State University.
- Extotoxnet, 2007b. Extension Toxicology Network: Pesticide Information Profiles–Metolachlor. Oregon State University.
- Frank, R., Clegg, B.S., Sherman, C., Chapman, N.D., 1990. Triazine and chloroacetamide herbicides in Sydenham River water and municipal drinking water, Dresden, Ontario, Canada, 1981–1987. *Arch. Environ. Contam. Toxicol.* 19, 319–324.
- Frank, R., Logan, L., Clegg, B.S., 1991. Pesticide and polychlorinated biphenyl residues in waters at the mouth of the Grand, Saugeen, and Thames Rivers, Ontario, Canada, 1986–1990. *Arch. Environ. Contam. Toxicol.* 21, 585–595.
- Gaynor, J.D., MacTavish, D.C., Labaj, A.B., 1998. Atrazine and metolachlor residues in Brookston CL following conventional and conservation tillage culture. *Chemosphere* 36, 3199–3210.
- Graziano, N., McGuire, M.J., Roberson, A., Adams, C., Jiang, H., Blute, N., 2006. 2004 National atrazine occurrence monitoring program using the Abraxis ELISA method. *Environ. Sci. Technol.* 40, 1163–1171.
- Hall, J.C., Van Deynze, J., Struger, J., Chan, C.H., 1993. Enzyme immunoassay based survey of precipitation and surface water for the presence of atrazine, metolachlor and 2,4-D. *J. Environ. Sci. Health B* 28, 577–597.
- Hayes, T.B., Haston, K., Tsui, M., Hoang, A., Haeffele, C., Vonk, A., 2002. Atrazine induced hermaphroditism at 0.1 ppb in American leopard frogs (*Rana pipiens*): laboratory and field evidence. *Environ. Health Perspect.* 111, 568–575.

- Kannan, K., Ridal, J., Struger, J., 2006. Pesticides in the Great Lakes. In: Hites, R.A. (Ed.), *Persistent Organic Pollutants in the Great Lakes*. Handbook of Environmental Chemistry. Springer-Verlag, Berlin, Heidelberg, pp. 151–199.
- Kramer, P.M., Martens, D., Forster, S., Ipolyi, I., Brunori, C., Morabito, R., 2007. How can immunochemical methods contribute to the implementation of the Water Framework Directive? *Anal. Bioanal. Chem.* 387, 1435–1448.
- Kurt-Karakus, P.B., Bidleman, T.F., Muir, D.C.G., Cagampan, S.J., Struger, J., Sverko, E., Small, J.M., Jantunen, L.M., 2008. Chiral current-use herbicides in Ontario streams. *Environ. Sci. Technol.* 42, 8452–8458.
- Kurt-Karakus, P.B., Muir, D.C., Bidleman, T.F., Small, J., Backus, S., Dove, A., 2010. Metolachlor and atrazine in the Great Lakes. *Environ. Sci. Technol.* 44, 4678–4684.
- Lawruk, T.S., Hottenstein, C.S., Fleeker, J.R., Hall, J.C., Herzog, D.P., Rubio, F.M., 1994. Quantification of 2,4-D and related chlorophenoxy herbicides by a magnetic particle-based ELISA. *B. Environ. Contam. Toxicol.* 52, 538–545.
- Lin, Y.J., Karuppiyah, M., Shaw, A., Gupta, G., 1999. Effect of simulated sunlight on atrazine and metolachlor toxicity of surface waters. *Ecotoxicol. Environ. Saf.* 43, 35–37.
- McGee, B., Berges, H., Beaton, D., 2010. Survey of Pesticide Use in Ontario, 2008: Estimates of Pesticides Used in Field Crops, Fruit and Vegetable Crops and Other Agricultural Crops. OMAFRA, Toronto, ON, Canada, p. 22.
- Osano, O., Admiraal, W., Otieno, D., 2002. Developmental disorders in embryos of the frog *Xenopus laevis* induced by chloroacetanilide herbicides and their degradation products. *Environ. Toxicol. Chem.* 21, 375–379.
- Phillips, P.J., Wall, G.R., Thurman, E.M., Eckhardt, D.A., VanHoesen, J., 1999. Metolachlor and its metabolites in tile drain and stream runoff in the Canajoharie Creek watershed. *Environ. Sci. Technol.* 33, 3531–3537.
- Rhoton, F., Shipitalo, M., Lindbo, D., 2002. Runoff and soil loss from midwestern and southeastern US silt loam soils as affected by tillage practice and soil organic matter content. *Soil Tillage Res.* 66, 1–11.
- Richards, R.P., Baker, D.B., 1993. Pesticide concentration patterns in agricultural drainage networks in the Lake Erie basin. *Environ. Toxicol. Chem.* 12, 13–26.
- Rubio, F.M., Itak, J.A., Scutellaro, A.M., Selisker, M.Y., Herzog, D.P., 1991. Performance characteristics of a novel magnetic-particle-based enzyme-linked immunosorbent assay for the quantitative analysis of atrazine and related triazines in water samples. *Food Agric. Immunol.* 3, 13–25.
- Schraer, S.M., Shaw, D.R., Boyette, M., Coupe, R.H., Thurman, E.M., 2000. Comparison of enzyme-linked immunosorbent assay and gas chromatography procedures for the detection of cyanazine and metolachlor in surface water samples. *J. Agric. Food Chem.* 48, 5881–5886.
- Selim, H.M., 2003. Retention and runoff losses of atrazine and metribuzin in soil. *J. Environ. Qual.* 32, 1058–1071.
- Statistics Canada, 2006. Census of Agriculture. Farms and Farm Operator Data Tables: Total Farm Area, Land Tenure and Land in Crops, by Province (Census of Agriculture, 1986–2006).
- Struger, J., Fletcher, T., 2007. Occurrence of lawn care and agricultural pesticides in the Don River and Humber River watersheds (1998–2002). *J. Great Lakes Res.* 33, 887–905.
- Struger, J., L'Italien, S., Sverko, E., 2004. In-use pesticide concentrations in surface waters of the Laurentian Great Lakes, 1994–2000. *J. Great Lakes Res.* 30, 435–450.
- Suzawa, M., Ingraham, H.A., 2008. The herbicide atrazine activates endocrine gene networks via non-steroidal NR5A nuclear receptors in fish and mammalian cells. *PLoS ONE* 3.
- Takacs, P., Martin, P.A., Struger, J., 2002. Pesticides in Ontario: A Critical Assessment of Potential Toxicity of Agricultural Products to Wildlife with Consideration for Endocrine Disruption, vol. 2. Triazine Herbicides, Glyphosate and Metolachlor Canadian Wildlife Service, Ontario Region, Burlington, Ontario, Canada.
- Thurman, E.M., Goolsby, D.A., Meyer, M.T., Kolpin, D.W., 1991. Herbicides in surface waters of the midwestern United States: the effect of spring flush. *Environ. Sci. Technol.* 25, 1794–1796.
- Waring, C.P., Moore, A., 2004. The effect of atrazine on Atlantic salmon (*Salmo salar*) smolts in fresh water and after sea water transfer. *Aquat. Toxicol.* 66, 93–104.
- Zaruk, D., Alae, M., Sverko, E., Comba, M., 1998. Occurrence of triazine herbicides and metolachlor in the Niagara River and other major tributaries draining into Lake Ontario. *Anal. Chim. Acta.* 376, 113–117.
- Zaruk, D., Comba, M., Struger, J., Young, S., 2001. Comparison of immunoassay with a conventional method for the determination of Diazinon® in surface waters. *Anal. Chim. Acta* 444, 163–168.